

**A Game Theoretical Approach to Stock Prices and Investor Strategies in Financial
Markets**

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Abstract

Game theory is a branch of mathematics that is concerned with interdependent decision-making of strategic agents. Game theory is applicable for situations where the decisions of the players are interdependent. Since it is in human nature to interact, game theory has been applied to many fields and most prominently economics. In such fields, game theory is utilized for simulating the relationship and developing strategic solutions for the actions of different players. Financial markets are most often unpredictable for an average investor and ultimately depend on the principle of supply and demand of different parties interacting in the market. This makes the utility of the investors in the financial market to depend on other investors' choices. Since the actions of investors and the market -the rest of the investors in the market- ultimately depend on each other, a game theoretical model could be appropriate to simulate this interaction. This paper aims to create a theory for a game that will stimulate utilities for stock prices and investor profiles. The simulation tested in this paper used historical data from different years 2008 and 2018 to create the model for the stocks. These years were especially chosen for adjusting the simulation for different financial market conditions. The paper finds that the specific game theoretical model proposed in this model fails to account for multiple factors in the financial market such as the unpredictable nature of investor decisions, therefore fails to accurately simulate real stock/index prices. This paper also ran the simulation for different time periods, varying levels of hypothetical volatility values and changing investor risk aversion levels and compared the results of these simulations with themselves and with real market conditions.

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1. Introduction

1.1. Background

Game theory is a branch of mathematics that deals with interdependent strategic decisions of rational decision makers. This strategic model is used for solving the complex decision-making processes behind real world scenarios in which the benefit of one player depends on the decision of another. With the assumption that the players are rational, one can apply this theory to many situations in real life ranging from interpersonal relationships to deals between governments. If the utility of an agent in the game can be defined using numbers through certain utility functions, game theory can be used for making strategic decisions (Rapoport & Chammah, 1965).

Financial markets are mediums where investments are traded. They have the role of connecting “surplus agents” such as households, government and companies to the other “deficit agents” that demand the investment. There are various types of financial markets, however, this paper will only focus on the stock markets and define the utility functions of investors accordingly (Pilbeam, 2018).

According to Pilbeam, the structures of financial markets affect the decisions of the decision-making agents participating in them by forming the “agents’ incentives to accumulate various types of physical and human capital, as well as other kinds of assets. This is true both because changes in market structure affect the perceived returns to various kinds of investments, and the risks associated with them” (2018). This nature of the financial markets causes the different agents such as investors and market makers to consider the others’ decisions while making the decisions of their own. Due to the decisions’ interdependence, it is possible to think of the financial markets as a game theoretical model.

1.2. Research Objectives

To create a simple two player game out of the stock market, I created a zero-sum model for financial markets using the assumptions of game theory and assuming investors and the market to be the two players of my game. My zero-sum model took the investors profit as the positive utility of the investor and takes the negative of the profit of the investor as the market utility, thus creating a zero-sum game. In each one-shot game, the market got to take two actions, and the player only takes one action. The actions of the market are the changing stock price according to geometric Brownian motion before and after the trading hours depending on the number of stocks the investor bought or sold during the day. The only action of the investor is to either buy or sell a certain number of stocks between the actions of the market during each trading day. With theoretical model, a Monte Carlo simulation using Geometric Brownian motion was created for the stock market. The simulation predicted stock prices and calculated the utility of the investor – first player – for each step. Later the final profits at the end of one year for 1000 simulations are plotted on a graph to measure the success of the investor.

This simulation made a variety of assumptions parallel to many games in game theory. Main assumptions of the theoretical model were that the investors are rational and have complete information about the working process of financial markets along with their utility in the financial markets. This model did not account for the behavioural game theory in terms of utilities though it presents the change in investors utility in different markets with varying levels of risk aversion. Another assumption this research made is regarding the number of actions the investors take during one trading day. The market is assumed to take two moves, and the investors are assumed to take one action in one trade day. There are two periods where the price of the exchange-traded fund or stock price changes, trading hours and non-trading hours to account for the off-market trades. The investor playing against the market could only buy, sell or do not take an action in one day during trading hours. In this research, I used prices for different stocks and times to account for the changing volatilities and drifts in the financial markets. Using the simulation, I first analysed the simulation's similarities and differences with real world stock data - imported from yfinance- and then I manipulated different variables to see their results in a market setting. The similarity between the simulation and the real-world stock data were obtained through a correlation coefficient calculation in the "Results" section. The variables that were analysed in this research are different periods for the financial market, the movement of different stocks/indexes and changing behaviour of investors in the market. In the hypothesis section, I presented my expectations from this research, later in the methodology section I described the process for the simulation. After methodology, in the results and discussion section, I presented my findings regarding the similarities and differences between my simulation

and real-world data. The manipulation of different variables was delved upon later in the discussions section.

The objective of this research was to provide a game theoretical framework for predicting stock prices using the interdependence of the decisions made by investors and the other sum of investors - the market- that they play against.

1.3. Significance

Game theory has found its applications in many different fields as an important tool to simulate interdependence decision making. The significance of this research is that it expanded the growing literature on game theory and simultaneously attempted to provide a different perspective and a model for the stock market. This research also aimed to bring light on the possible consequences of different strategies the investors might follow in a stock market. With this aspect, this model also defines some strategies that were assumed to be more “risk averse” and saw how the behavioural differences between players affected their earnings in the stock market.

2. Literature Review

2.1. Game Theory

Game theory has been the focus of economics in many ways such as oligopolies and tragedy of the commons. Game theory is famously known for being a mathematical model for strategic interactions. Since, game theory aims to replicate strategic interactions through mathematics, it must make some assumptions. In game theory the main assumptions are:

- There are finite number of players in a game.
- The players act rationally meaning their actions are consistent with their previous actions.

- The players act to maximize their utilities according to the given utility functions.

Utility functions are used for defining the benefit one player will get because of the strategies followed in a game. Defining these utility functions can sometimes be problematic as well for many cases in real life, players do not have enough knowledge to determine their utilities and act accordingly. In this research paper, the game used is a two-player zero-sum game and the players were assumed to act rationally with complete information. Academic interest in game theory started with John Van Neumann around 1928. In 1928, Van Neumann proved the minimax theorem which states that for every two-investor zero-sum game^[1], there is a saddle point^[2]. Zero-sum games assumed ultimate conflict of interests between agents, however.

Another important milestone in the history of game theory was the formation of a solution concept in non-zero-sum games, Nash equilibrium, that was pioneered by Nobel laureate John Nash (Rapoport & Chammah, 1965). Although the concept of non-zero-sum games better encompasses the events happening in real life, the solutions for such games are more complex and provide less efficient and definitive solutions than solutions of zero-sum games. As history progressed and game theory became a recognized way to create models for interdependent decision-making interactions, game theory became increasingly integrated in the literature of many sciences ranging from biology to economics.

2.2. Literature on Stock Price Prediction

There had been many attempts to come up with simulation methods for the stock market as it is often unpredictable by many. One of the popular methods was to use Monte Carlo simulation with geometric Brownian motion. Monte Carlo simulation is an attempt to predict the

future through simulating the many possible outcomes. The simulation produces a distribution of possible random outcomes for the researcher to interpret. Monte Carlo simulation was named after the famous gambling destination in Monaco because chance and random outcomes are central to random chance in gambling as well. The technique was developed by Stanislaw Ulam and later John von Neumann joined him to improve the technique (Harper).

2.3. Past Game Theory Applications in Financial Markets

The idea of incorporating game theory into financial markets is not new. One of the efforts to do so was through adjusting the payoff functions of the decision makers in the financial markets to the conditions in the market. Assume that there are different companies competing in a market. According to Hannafey, each company's decision to adjust prices or trading volume was like choosing a strategy in Cournot or Bertrand games and the correlation patterns in difference vectors vs. volume suggest reaction functions of investors to other investors actions in the financial markets (Hannafey, 2021). Cournot and Bertrand games are two types of game theories formed for attempting to explain duopolies. Cournot simulates scenarios in which players compete by changing their output, whereas in Bertrand games, firms change their prices to compete. In another paper, Evangelista claimed that the "price formation of asset prices [as] a process constituted as a result of the interaction of traders through the limit order book" (2022). Like the model used in this research, the price formation was claimed to be related to "the correlation between an incoming order (to buy or to sell) and the subsequent price change" (2022), though the model did not use a two-investor game. Instead, the model takes the traders as investors and the market as "a single venue in which infinitely many agents trade a single asset"

(Evangelista, 2022). In contrast with the papers, this paper addressed the financial market as a two-investor zero-sum game.

3. Methodology

3.1. Collection and Processing of Data

The analysis started by importing data for different stocks for different years from Yahoo Finance (yfinance) through a Python program. The import command from yfinance allows the download of historical stock open and close prices, volume and various market indicators. Using the data I collected from Yahoo Finance, I calibrated my model for the drift and volatility values of the stock price for 2018. The game was made up of three rounds and 252 iterations, one iteration for each trading day in a year. The first round belonged to the market, and it was a step to determine the stock price for before the trading hours start. In this step, the stock price changed according to geometric Brownian motion with a standard normal variable (ϵ) with normal distribution. In the second movement of the market, this variable was changed to adjust for the market conditions. After this step, depending on the increase or decrease in the stock price, the investors either buy or sell the stock or did nothing as their action.

The investor could take three actions. If the stock price increased above a certain percentage (standard percentage increase or decrease for the initial trial is 10%) from the previous price, the investor played “up” meaning the investor was more likely to sell more of the stock. If the stock price decreased below a certain percentage from the previous price, the investor played “down,” meaning the investor was more likely to buy more of the stock. If the stock price change was not above or below the thresholds, the investor was equally likely to buy or sell the stock.

After the investor buy or sells the stock, the market took an action according to the investor's action in the previous round. The logic of the movement of the stock price was like that of the first round of the market. However, depending on the net number of stocks sold or bought by the investor, the random variable (ϵ) changed. If the investor bought more stocks, this led the market to be more likely to increase the stock price by choosing the value of ϵ from a left skewed number distribution. If the investor sells more stocks, this will lead the market to be more likely to decrease the stock price by choosing the value of ϵ from a right skewed number distribution. This process was similar to stock price changes according to principles of supply and demand in the stock market.

After the last movement of the market, one round was completed, and the cycle began from the beginning with the market acting according to the geometric Brownian motion with normally distributed ϵ value. This cycle was repeated for 252 for every game to simulate the number of trading days in a year. One game of 252 iterations was repeated for 1000 times using Monte Carlo simulation technique. The profit of the investor was assumed to be:

$$Utility = Stock\ Price \times Number\ of\ Stocks + Cash\ in\ Hand$$

$$Profit = Utility\ Final - Utility\ Initial$$

The investor started with 10000 dollars and 10 stocks, each at value $S_0 = 10$ dollars. The profit for the market investor was assumed to be the negative of that of investors to simplify the analysis of the game. Both stock price data and the difference between the stock price and the

real-world stock data was recorded in a list for the comparison between rounds and the final graphs. A distribution of the coefficient correlations was formed to see the correlation between the real-world data and the prices modeled by the model created in this research. Google Colab^[3] was used for the production and the execution of the Python code for this model and data was imported from Yahoo Finance.

3.2. Fundamental Mathematical Models

In this section, the mathematical foundations of the calculations were explained. There are two main mathematical methods that were used in the simulation, Monte Carlo simulation and geometric Brownian motion.

3.2.1. Geometric Brownian Motion

Geometric Brownian motion is a Markov process. This means that the stock price follows a “random walk” which is consistent with the weak form of the efficient market hypothesis. In other terms, the next price movement in the stock/index is conditionally independent of past price movements in the stock market.

The formula for GBM is:

$$\frac{\Delta S}{S} = \mu \Delta t + \sigma \epsilon \sqrt{\Delta t}$$

Where:

$$S = \text{the stock price}$$

$$\Delta S = \text{the change in the stock price}$$

$$\mu = \text{the expected return}$$

σ = the standard deviation of returns (volatility)

ϵ = the random variable

Δt = the elapsed time period

If the formula is rearranged, the equation yields:

$$\Delta S = S \times (\mu \Delta t + \sigma \epsilon \sqrt{\Delta t})$$

(Harper)

3.2.2. Monte Carlo Simulation

Monte Carlo simulation is a mathematical method for estimating the potential outcomes of an uncertain event. Monte Carlo Simulations have assessed the impact of risk in many real-life scenarios, such as in artificial intelligence, stock prices, sales forecasting, project management, and pricing. Monte Carlo simulation helps to simulate a potential outcome through repeated computation of possible outcomes using a model. The model this paper used in the Monte Carlo simulation is geometric Brownian motion (GBM). For each simulation run among the many in the Monte Carlo simulation, a certain value of drift and volatility were used for predicting the future stock price and the game is played for 252 iterations. These volatility and drift values were obtained through the historical data from Yahoo Finance. Then the average payoff for all the games played during the simulation was calculated while also plotting the distribution of games.

4. Results and Discussion

In the Result and Discussion section, first the real-world stock prices and the stock prices obtained from the simulation was compared using the correlation coefficient between them. This comparison was done to assess the accuracy of the game theory simulation for predicting stock

prices in an actual market. The accuracy was tested for different stock price information – the opening and closing prices – and different time periods.

4.1. Assessing the Similarities and Differences of the Simulation with Real World Stock Prices

In this part, the similarities between the real stock prices and the ones calculated in the simulation were calculated through a correlation coefficient calculation taking the different corresponding stock values for the same number of time iterations (one of the trading days in 252 days).

4.1.1. Results of the Simulation for the First Stock and Comparison with the real Stock Value

In this part the correlation between the resulting stock prices from the simulation and real-world data were addressed after the discussion about the profitability of the investor in the simulation. The stock prices were simulated using the game theory model that uses geometric Brownian motion and Monte Carlo simulation mentioned in the methodology section. The profit distribution for the investor was given as Figure 1 for a Monte Carlo simulation for 1000 games. The investor in this game was programmed to be more likely to sell for a 10% increase in the stock price and to be more likely to buy for a 10% decrease in the stock price. The market acted as specified in the methodology section.

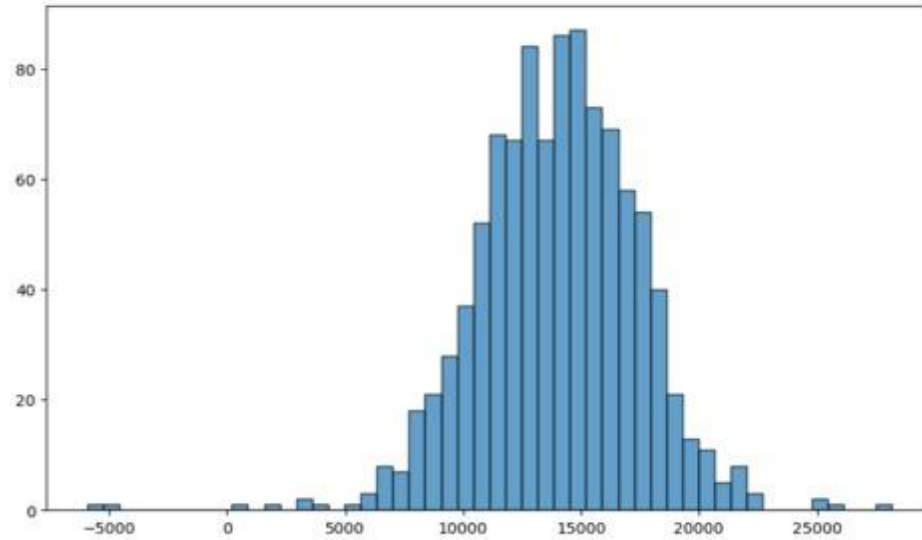


Figure 1

(Distribution of Profits of the Investor in the Game)

According to the distribution (Figure 1), most of the time investors will earn profit. This makes sense as the investor is coded for selling the stock when the stock price increases and buying new stocks when the stock price decreases. The investor played for creating profit through the exchange of stocks. The mean of the profits according to this run is 13992.482462399494 dollars for the investor. This makes the profit of the market the opposite-13992.482462399494dollars. This simulation assumes that the investor earns the money by using the utility of the market. Therefore, this interaction is modelled as a zero-sum game.

Figure 1 above shows that the profits of an investor participating in this game has an approximately normal distribution. This return is similar to the fat-tailed normal distribution for rates of returns in the financial market. The distribution of the rate of returns in a financial market is called “fat-tailed” due to the more extreme values having more probability to occur

than in a classic normal distribution. Since this model makes simplifying assumptions for the actions for investors and does not account for the large jumps that may occur in the stock price due to extraordinary events independent from the stock market that affects the market, the distribution is more similar to a normal distribution that one may encounter after doing a Monte Carlo simulation with geometric Brownian motion. The imperfections in the normal distribution can be explained through the different action options of the markets (up, down) when the investor has a net positive or negative number of stocks bought.

The model was calibrated with the data from 2018 volatility and drift values for S&P index, therefore the accuracy of the period should be compared with this time period with this specific index. To determine the correlation between each stock value for each day in this model, I used the stock data from each simulation data and the real-world data to calculate their correlation coefficient and obtained the Figure 2 distribution below for the Monte Carlo simulation of 1000 repetitions of the game.

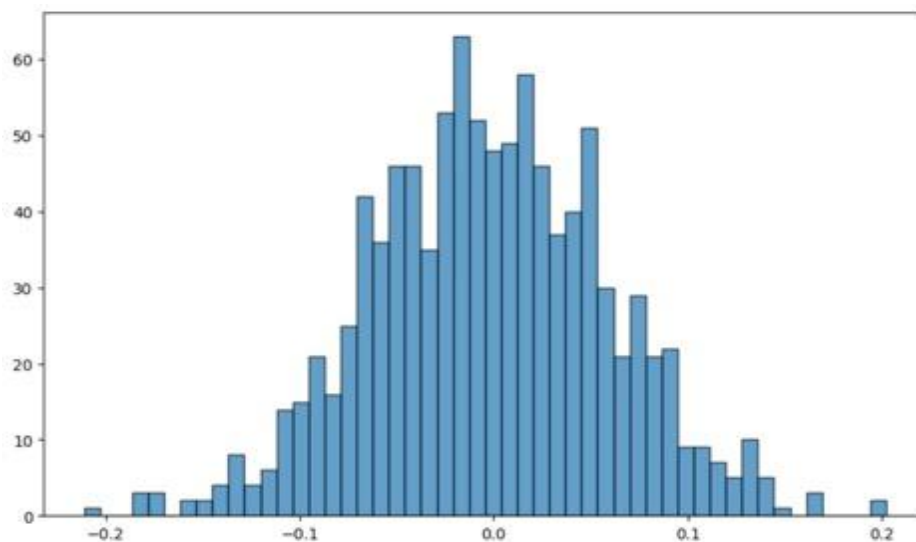


Figure 2

(Correlation Coefficient Distribution of 1000 Iterations of the Game with Opening Prices
of Real S&P Index Data)

Figure 2 clearly shows that many of the correlation coefficient values of the 1000 trials in the Monte Carlo simulation are close to 0. This indicates that there is no significant correlation between the stock prices calculated by the model and the stock prices in the real world. This phenomenon is also explained by the mean of the correlation coefficient values that is 0.001163635224447533. This model worked with a variety of assumptions, which ultimately led it to lose correlation with the progress of the stock market in 2018 for S&P index. In the real world, there are thousands of investors that buy millions of stocks and this small simulation assumed that the investors could only buy or sell a maximum of 400 stocks in one trading day. Furthermore, the market is assumed to only change by Geometric Brownian Motion that fails to account for the large jumps that may happen in the stock market during extreme periods. Furthermore, the lack of opportunities for the actions for investors and the stock market may also have caused the discrepancy between the stock market values for this model and the real-world stock market values. Although one might assert that this phenomenon might be the case for this index, S&P, only, through the trials with different stocks and indexes, similar results have been obtained. Such stock values ran through this simulation were as follows: Tesla, Amazon, Microsoft, Nvidia, Apple and Google. The trials led to the following means for correlation coefficients respectively: 0.004233737467418467, -0.0006410867206942041, 0.003882588477902594, 0.0005003184032030403, -0.0061128987358224605, -0.0006594418350023

4.1.2. Assessing the Accuracy with Closing Stock Data

Since the results for the accuracy of the model for the opening prices were not aligning with the real-world opening prices, in this section an attempt to simulate the closing prices for the stock is made. The opening prices of the day might have been more related to the events happening out of the trading hours and therefore might have been unpredictable by the model that depends on player decision making during market hours. However, regardless of the utilization of closing or opening data in the day, the results – as demonstrated by Figure 3- were similar and the average of the correlation coefficient for the Monte Carlo simulation was around the same as the previous trial with opening prices of the stock, that is 0.0029288077262252465. The small amount of correlation between stock prices indicate that the prices the simulation yielded were far off from the real index prices for S&P in 2018.

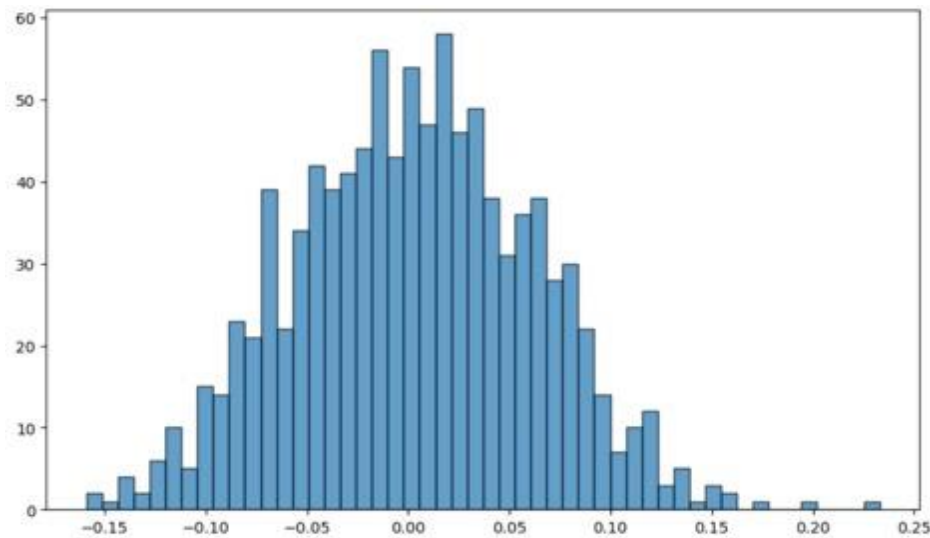


Figure 3

(Correlation Coefficient Distribution of 1000 Iterations of the Game with Closing Prices
of Real S&P Index Data)

4.1.3. Comparison of the Predictive Power of the Simulation Across Different Times

Since the stock prediction model was not successful to predict the stock prices for different stocks and its inaccuracy was not caused by the fact that the comparison was made with the opening or closing prices, this part compared the stock prices predicted by the model and the actual stock prices from different time periods in history. Since 2018 was not particularly famous for its financial crisis, this section used 2008 to see whether the predictive ability of the model increases during times of financial crisis. Here is the outcome of the simulation for index price data calibrated by 2008 index data of S&P:

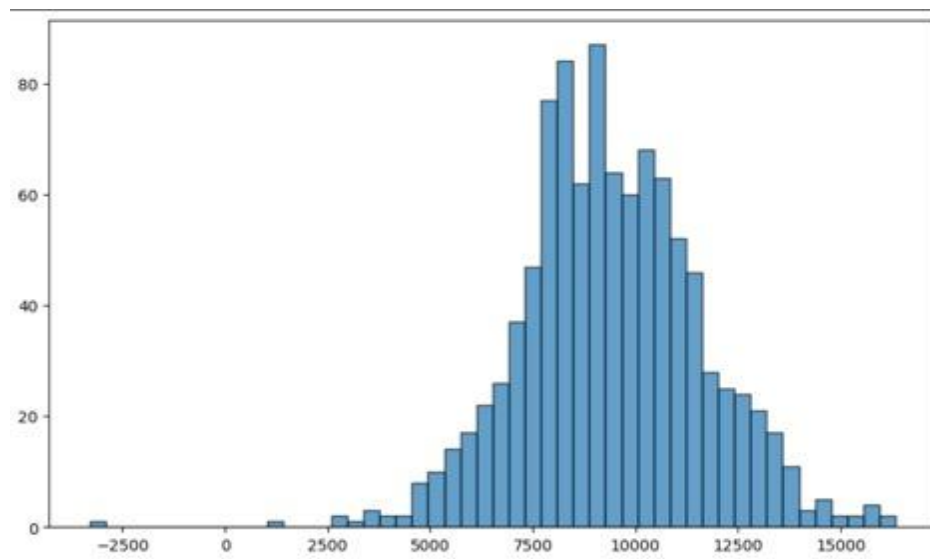


Figure 4

(Distribution of Profits of the Investor in the Game for 2008 Volatility)

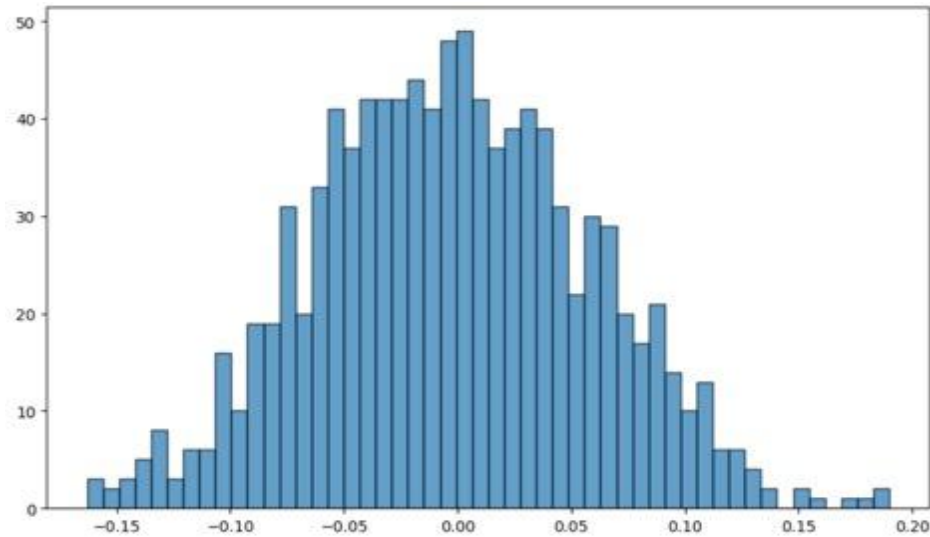


Figure 5

(Correlation Coefficient Distribution of 1000 Iterations of the Game with Opening Prices
of Real S&P Index Data for 2008 Data)

As seen from Figure 5, the correlation coefficient of this trial is not significantly different from the previous trials - -0.0027971833200927275 -. One important change was about the profit of the player. The profit for the player in 2018 fluctuated around 140000 in the first trial. Changing the time of the data collection period to a more unpredictable and volatile period such as the 2008 financial crisis led the investor to earn less profit, resulting in the player earning only 9395.907267830216 dollars of profit instead as shown on Figure 4. This indicates that the profit of the investor was susceptible to changes in volatility values and the stability of the market.

5. Additional Discussion

In this part, I manipulated different variables in the simulation to see their effects in the stock market. The variables that were changed in this section for fine tuning the model are as follows:

- I. Investor's Risk Aversion in a Market with Low Volatility
- II. Investor's Risk Aversion in a Market with High Volatility

5.1. Changing the Investor's Risk Aversion in a Market with Low Volatility

In this simulation, hypothetical values for the volatility were used for testing the hypothesis that there will be a change in investor profitability with changing levels or risk aversion. The volatility and drift values were not adjusted for the real-world volatility and drift values. The volatility that will be used for testing for low volatility will be 5%. The risk aversion for the investor will be adjusted by changing the parameters in its actions. For example, the investor was more likely to buy stocks when the value of the stock was less than 90% of its value in the previous period. In contrast, the investor was more likely to sell stocks when the value of the stock was greater than 110% of its value in the previous period. To decrease the risk aversion of players, these parameters will get closer to 100 so that the investor can start to take more rapid actions even with a slight change of price in the stock. To increase the risk aversion of the investors in the market, the rate of change necessary in the stock price for the players to take significant action will be increased so the investors are more hesitant to buy or sell stocks. All the calibration data used for this model for this run is from 2018.

The simulation was run for four types of players and the maximum number of stocks a player can buy or sell in a day is 400.

1- Type 1 (Standard Player) –

- a. The player is more likely to play “sell” – be more likely to sell stocks – if there is a %10 increase in the stock price.

- b. The player is more likely to play “buy” – be more likely to sell stocks – if there is a %10 decrease in the stock price.
- c. The player buys or sells a random number of stocks between 0 and 250 if the change in the stock price does not exceed the threshold of 10%.

2- Type 2 (Risk Averse but random actor)

- a. The player is more likely to play “sell” – be more likely to sell stocks – if there is a %30 increase in the stock price.
- b. The player is more likely to play “buy” – be more likely to sell stocks – if there is a %30 decrease in the stock price.
- c. The player buys or sells a random number of stocks between 0 and 250 if the change in the stock price does not exceed the threshold of 30%.

3- Type 3 (Risk Taking Player)

- a. The player is more likely to play “sell” – be more likely to sell stocks – if there is a %1 increase in the stock price.
- b. The player is more likely to play “buy” – be more likely to sell stocks – if there is a %1 decrease in the stock price.
- c. The player buys or sells a random number of stocks between 0 and 250 if the change in the stock price does not exceed the threshold of 1%.

4- Type 4 (Risk Averse and Inactive Player)

- a. The player is more likely to play “sell” – be more likely to sell stocks – if there is a %30 increase in the stock price.

- b. The player is more likely to play “buy” – be more likely to sell stocks – if there is a %30 decrease in the stock price.
- c. The player does not buy or sell at all if the difference in the stock price is below the threshold of %30.

Initially, the simulation is run with the standard level of risk aversion (Type 1) for this research as stated above. This run leads the following results in Figure 6.

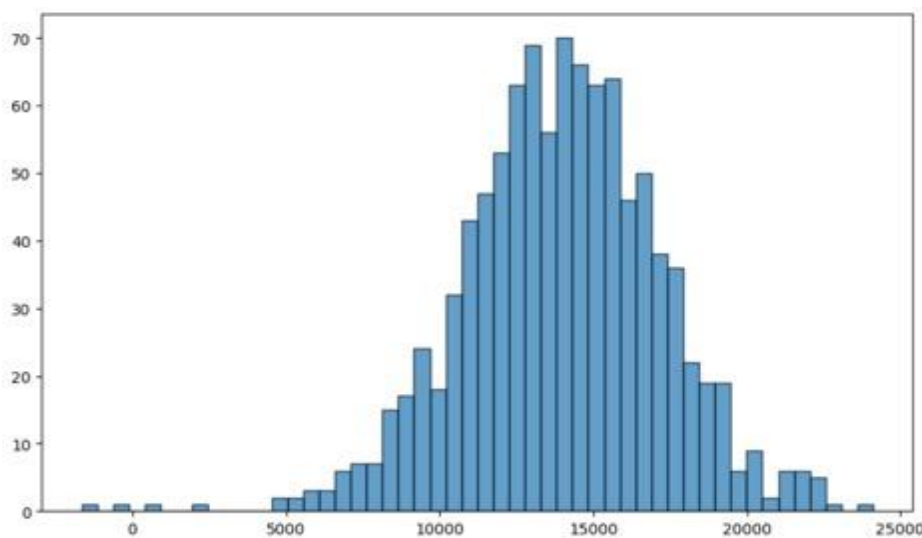


Figure 6

(Distribution of Profits of the Investor in the Game for Player Type 1 in Low Volatility)

Type 1 player was programmed to be moderately aware of the progress of the stock price and as a result it earned 13958.408276878961 dollars of profit on average after 252 trading days.

Secondly the simulation was run for Type 2 player. The risk averse player was programmed to act less susceptible to the changes in stock prices. The profit distribution of this player was demonstrated on the distribution below (Figure 7).

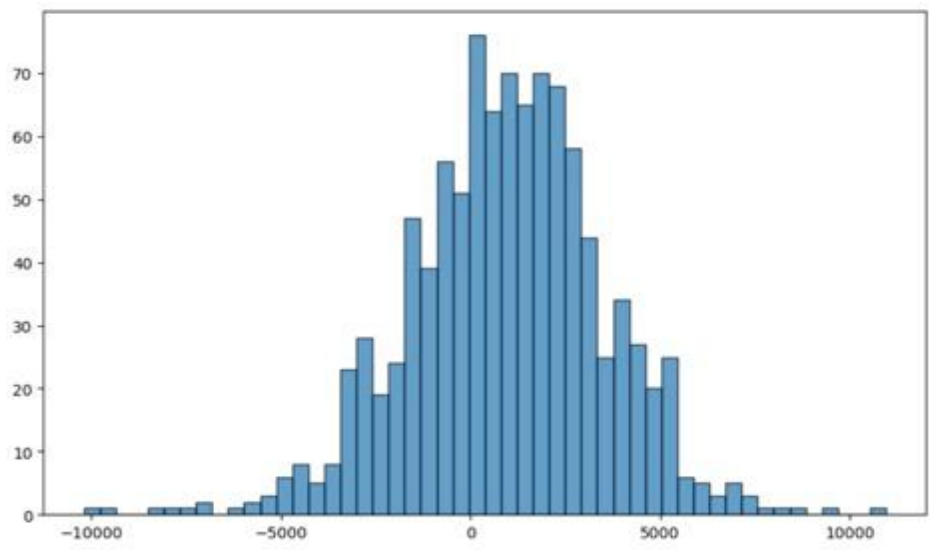


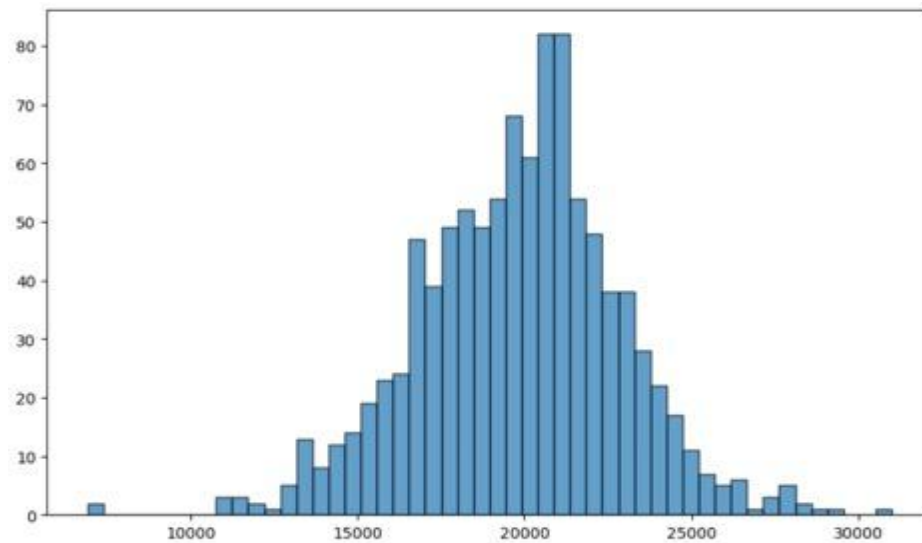
Figure 7

(Distribution of Profits of the Investor in the Game for Player Type 2 in Low Volatility)

According to the stock market depicted in Figure 7, the player to act less susceptible to the stock prices results in less profit for the player, that is 935.6484120438801 dollars of profit on average. Therefore, according to this model, it is important for players to be constantly aware of the market and adjust their strategies accordingly rather than acting randomly.

Next, the simulation was run with a player that is more informed about the market and more susceptible to change and adjusts its strategy for the changing prices of the stocks, player

Type 3. This player will change its strategy to buy or sell even with a 1% increase or decrease in



the stock price.

Figure 8

(Distribution of Profits of the Investor in the Game for Player Type 3 in Low Volatility)

This player strategy that adjusts even for the smallest change in the stock price performed even better than the previous two models and gained 19813.648456732033 dollars of profit on average (Figure 8). This further emphasizes the claim that a strategy that better adjusts to the conditions of the market will be performing better than the rest in the market.

Lastly, the simulation is ran with a player that does not randomly buy or sell when the stock price does not change by more than 30%, player Type 4. Unlike the second player tested for this section, instead of randomly buying and selling for changes less than the threshold in the stock price, this player does not sell or buy at all.

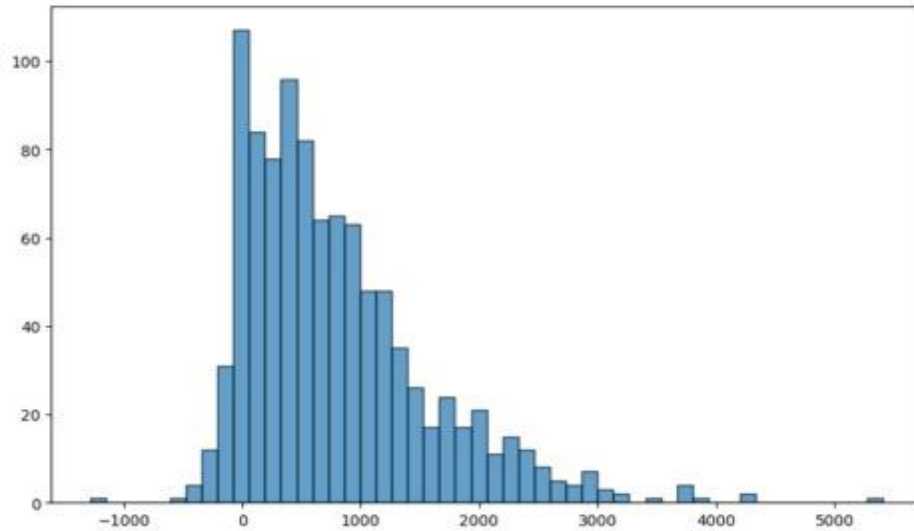


Figure 9

(Distribution of Profits of the Investor in the Game for Player Type 4 in Low Volatility)

Although this last player also performed almost as unsuccessful as the first player that played randomly on average – earning 790.631536973898 dollar of profit on average -, as demonstrated on Figure 9, the distribution of its earnings had a left skew meaning that more of its payoffs were lower than that of the random player. This indicates that a player whose buying strategy is less dependent on the stock prices will still perform better than a player who does not act if the change in the stock price does not change more than the threshold. The last player's profit was closest to zero among all of the other types of players. This was the case because for many turns of the game, this player – Type 4 – did not play at all. According to Figure 10 below, the number of times the player actually took action in a 252 day game did not exceed 13 times per game.

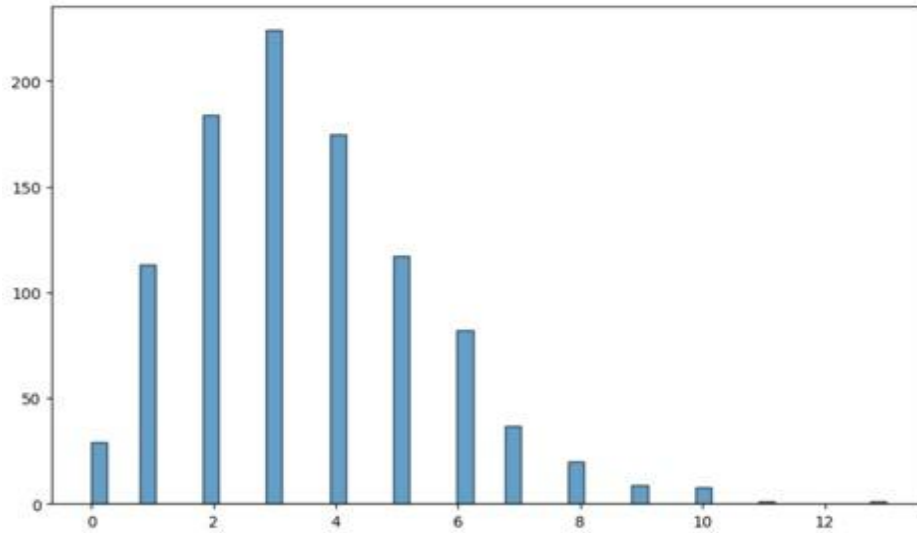


Figure 10

(Distribution of the Number of Times Player 4 Made in 252 Trading Days)

5.2. Changing the Investor's Risk Aversion in a Market with High Volatility

To test the accuracy of the hypothesis that suggests a player who better adjusts its strategy to market conditions will perform better in general, the hypothesis was tested under high volatility conditions as well. In this simulation, hypothetical values for the volatility were used for testing the hypothesis. The volatility and drift values were not adjusted for the real-world volatility and drift values. The volatility used for testing for high volatility was 80%. The risk aversion for the investor was adjusted by changing the parameters in its actions. For example, the investor was more likely to buy stocks when the value of the stock was less than 90% of its value in the previous period. In contrast, the investor was more likely to sell stocks when the value of the stock was greater than 110% of its value in the previous period. To decrease the risk aversion of players, these parameters will get closer to 100 so that the investor starts to take more rapid actions even with a slight change of price in the stock. To increase the risk aversion of the

investors in the market, the rate of change necessary in the stock price for the players to take significant action will be increased so the investors are more hesitant to buy or sell stocks.

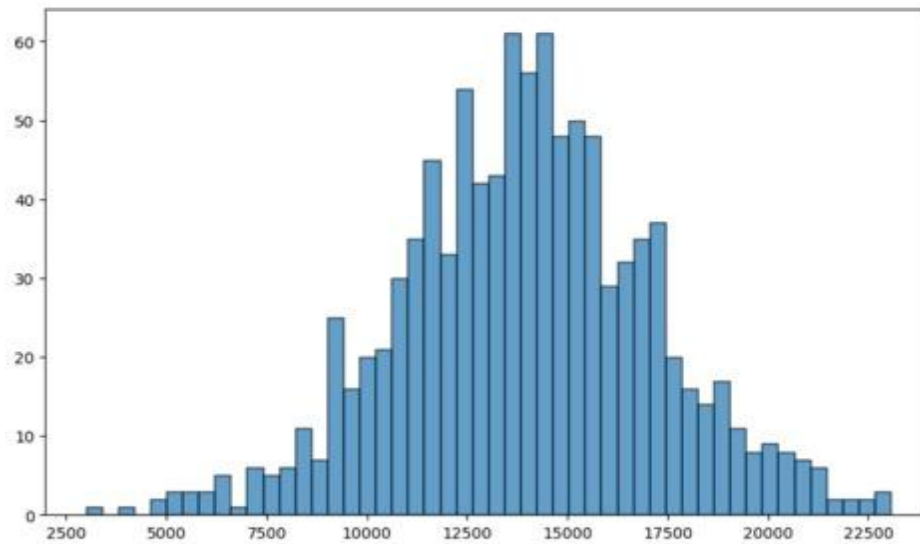


Figure 11

(Distribution of Profits of the Investor in the Game for Player Type 1 in High Volatility)

The standard player – Type 1- , on average, performed similar in a stock market with a high volatility, acquiring 13952.37440246342 dollars of profit. The player is still earning positive returns and the distribution of the Monte Carlo simulation of its returns is a normal distribution as shown on Figure 11.

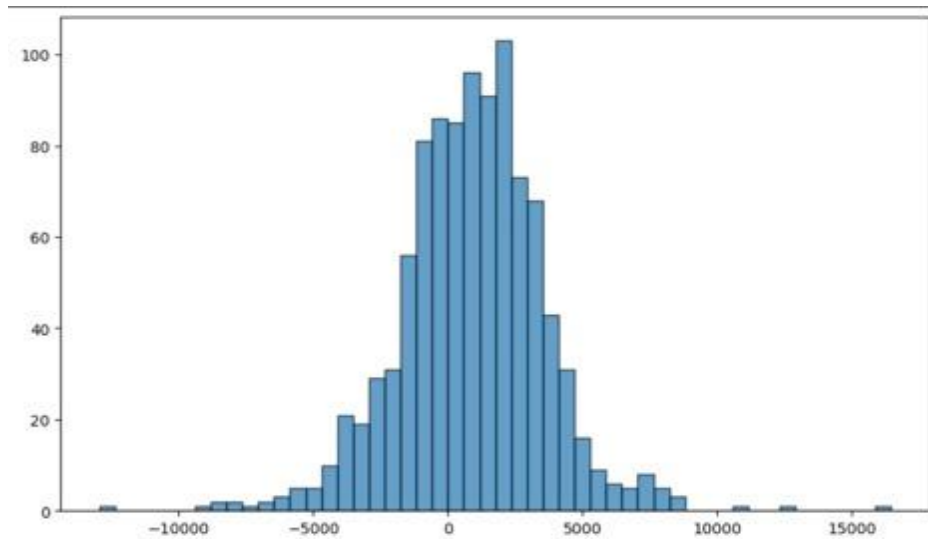


Figure 12

(Distribution of Profits of the Investor in the Game for Player Type 2 in High Volatility)

As seen from Figure 12, there is not a significant difference between the performance of Type 2 and its performance for low volatility. The player 2's profit distribution is still normally distributed, and the player earns an average of 884.3428465363412 dollars as demonstrated on Figure 12.

The simulation is again run with the player that responds quickly to market conditions – Type 3. Like previous market conditions, this player again earns positive profit – an average profit of 20054.60387436459 - and the profit distribution of the Monte Carlo simulation for its profits is a normal distribution as demonstrated on Figure 13.

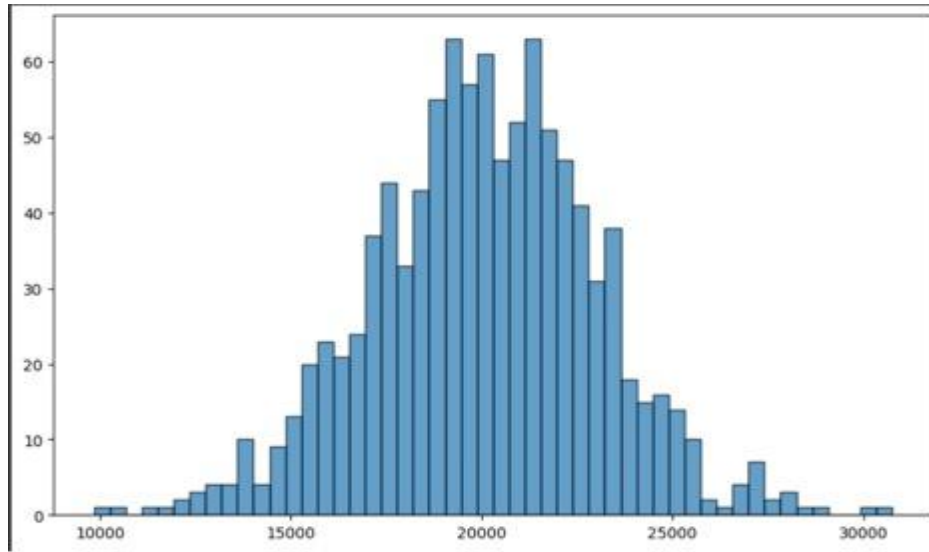


Figure 13

(Distribution of Profits of the Investor in the Game for Player Type 3 in High Volatility)

Finally, the simulation is run once again with the nonresponsive player (Type 4) that only buys or sells a random amount of stock prices when the stock price changes above or below a threshold. This threshold is once again the same as the second player (Type 2), that is 30% change in the stock price.

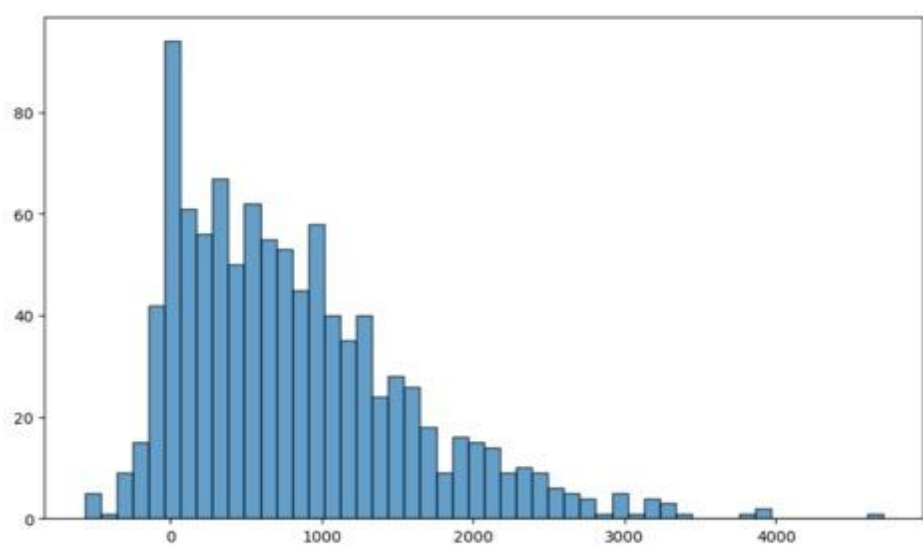


Figure 14

(Distribution of Profits of the Investor in the Game for Player Type 4 in High Volatility)

Seeing the mean profits and the distributions of the profits are similar for low volatility conditions for all player – an average profit of 813.6925133663228 dollars as depicted on Figure 14 -, it will be suitable to make the conclusion that the player is more susceptible to stock price changes and better adjusts its strategy (Type 3) accordingly will consistently get higher profits than the players who are unable to act according to the market conditions.

6. Limitations of the Model and Conclusion

There were certain limitations for these simulations. The model used geometric Brownian motion to predict stock prices, which makes assumptions for a perfectly efficient market and rate of returns. The game assumed that the investor can do one action per day (iteration). Although the model aimed to account for the different changes in open market hours and off-market hours by simulating geometric Brownian motion for two times an iteration, this may still not account for the volatility in the real world. Furthermore, the players were expected to act rationally and act according to the rules that were defined for their strategies, which is not the case in the real world. This assumption can be addressed through adding noise to the simulation in further research. Furthermore, this research did not encompass all the different strategies one player might follow, there can be more complex strategies developed and tried out with this simulation. Therefore, another research that encompasses such trials may help towards adding to the literature on this simulation.

The game theoretical model in this research paper unfortunately failed to create a simulation for the real-world stock market values as demonstrated by the low correlation

coefficients for different times and different stock prices. However, as demonstrated in the additional discussions part of this paper, this paper showed that players who were quicker to adjust their strategies to the changing stock market will, on average, get more profit from their investments. These results can demonstrate the working principle behind high-frequency trading in which is a fast algorithmic automated trading system.

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[1] A game is called “zero-sum” if the utilities for every strategy combination for investor lead to utilities that sum up to 0.

[2] A point is a saddle point of a game if the payoff of a cell in a matrix is the smallest of its row and maximum of its column.

[3] Google Colab is a coding platform and a tool for Python development.